PAPER_528 — UQFF_comp Spectral Compression Eigenvalue Stability

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

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PAPER_528 — UQFF_comp Spectral Compression Eigenvalue Stability

Author: Daniel T. Murphy **Framework:** Star-Magic / UQFF **Version:** v5.02 **Date:** 2026-03-25 **Session:** 142 — grok_share_2515709ed.txt **CP4 Class:** UQFFCompSpectralMatrixEigenvalueCalculator (#123) **Quality Score (QS):** 5 / 5

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§1 — Overview

UQFF_comp is the spectral compression tensor introduced in Session 141 (PAPER_522) to encode the Universal Spectrum's 1/3 stable / 2/3 destructive partition into a matrix formalism. This paper proves its eigenvalue stability and derives the boundedness condition.

$$\text{UQFF\_comp} = \begin{pmatrix} P/3 & 0 & 0 \\ 0 & P/3 & 0 \\ 0 & 0 & 2P/3 \end{pmatrix}$$

where $P = P_\text{order}$ from PAPER_527 (Pymander Sphere).

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§2 — Eigenvalue Analysis

Eigenvalue	Value	Sector	Multiplicity
λ_stable	$P/3$	Stable (our existence)	2
λ_destruct	$2P/3$	Destructive	1

Key relationships:

$$\lambda_\text{destruct} = 2\,\lambda_\text{stable}$$

$$\text{Tr}(\text{UQFF\_comp}) = \frac{4P}{3}, \qquad \det(\text{UQFF\_comp}) = \frac{2P^3}{27}$$

$$\|\text{UQFF\_comp}\|_F = P\sqrt{\frac{2}{3}}, \qquad \rho(\text{UQFF\_comp}) = \frac{2P}{3}$$

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§3 — Boundedness Theorem

Theorem: UQFF_comp is spectrally bounded ($\rho \leq 1$) if and only if $P \leq 3/2$.

Proof: $$\rho = \frac{2P}{3} \leq 1 \iff P \leq \frac{3}{2}$$

Since $P = e^{-E/F_\text{max}} / Z$ and $Z = \mathrm{Li}_{26}([SSq]) \approx 0.570 > 0$, we have $P \leq 1/Z \approx 1.75$.

For all physical systems where $E \geq F_\text{max} \ln(1/Z) \approx 0.562 F_\text{max}$: $$P \leq \frac{1}{Z} \cdot e^{-E/F_\text{max}} \leq 1 < \frac{3}{2}$$

$$\boxed{\text{UQFF\_comp is bounded for all physical systems with } E \geq E_\text{min}}$$

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§4 — UQFF Number Systems Integration (PAPER_429)

Vacuum Density Series (VDS)

The partition function $Z = \mathrm{Li}_{26}([SSq]) \approx 0.570$ directly normalises P, ensuring the spectral radius remains below 1 for physically realised entropy values. Without VDS, the eigenvalue stability theorem would require an arbitrary normalisation constant.

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§5 — Connection to Session 141 Physics

Session 141 concept	UQFF_comp encoding
A_stable fraction (1/3)	$\lambda_\text{stable} = P/3$
D_repel fraction (2/3)	$\lambda_\text{destruct} = 2P/3$
Off-diagonal couplings	Off-diagonal entries = 0 (diagonal in this basis)
Spectral tensor bounded	$\rho \leq 1$ proved via VDS

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§6 — Available Equations

Equation	Description		
$\text{UQFF\_comp} = \text{diag}(P/3, P/3, 2P/3)$	Full matrix form		
$\rho(\text{UQFF\_comp }) = 2P/3$	Spectral radius		
$\	\text{UQFF_comp}\	_F = P\sqrt{2/3}$	Frobenius norm
$\det = 2P^3/27$	Determinant		
$P_\text{max} = 3/2$ — boundedness limit	Critical probability		
$E_\text{min} = F_\text{max} \ln(1/Z)$	Minimum entropy for stability		

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§7 — Simulation Set

1. **Eigenvalue vs Prob_order sweep:** Plot λ_stable and λ_destruct as functions of P over $[0, 1.5]$ — observe spectral radius crossing at $P = 3/2$.
2. **Stability boundary:** Map E/F_max vs ρ for $[SSq] \in [0.1, 1.0]$.

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§8 — CP4 Calculator Output

```
calc = UQFFCompSpectralMatrixEigenvalueCalculator()
result = calc.compute(dataset={&#39;Prob_order&#39;: 1e-5})
# result[&#39;lam_stable&#39;]   — λ_min = P/3
# result[&#39;lam_destruct&#39;] — λ_max = 2P/3
# result[&#39;bounded&#39;]      — True if P ≤ 1
# result[&#39;det&#39;]          — 2P³/27
```

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§9 — References

• PAPER_429: Three New UQFF Number Systems (VDS / DVP / BH)
• PAPER_521: Universal Spectrum Spectral Divisions
• PAPER_522: DPM as Quantum Frequency Driver / UQFF_comp tensor introduction
• PAPER_527: Pymander Sphere (defines P_order)
• grok_share_2515709ed.txt: BigBangHypergraphTheory Millennium proof set